WORKSHEET — UNIT III

NumPy and Pandas

Course: Data Science with Python **Unit:** III (12 Hrs)

Name: _____ **Roll No:** _____ **Date:** _____

Topics covered: NumPy arrays • Series & DataFrame • Statistics • Missing data • Hierarchical index

Section 1 — Key Concepts (Quick Revision)

Term	Simple Meaning	Example
NumPy	Library for fast number work using arrays.	`import numpy as np`
ndarray	NumPy's array. All items have the same type.	`np.array([1, 2, 3])`
Shape	The number of rows and columns of an array.	`arr.shape → (5, 3)`
Vectorization	Work on a whole array at once, with no loop.	`arr * 2`
Broadcasting	NumPy matches different shapes automatically.	`arr + 10`
Pandas	Library for working with table data.	`import pandas as pd`
Series	One column of data with labels (1-D).	`pd.Series([10, 20, 30])`
DataFrame	A table with rows and columns (2-D).	`pd.DataFrame(data)`
Descriptive statistics	Summary numbers such as mean, median and std.	`df.describe()`
Missing data (NaN)	Empty values in the data. Find them first.	`df.isnull().sum()`
dropna / fillna	dropna removes rows with NaN. fillna replaces NaN.	`df.fillna(0)`
Hierarchical index	More than one level of row labels (MultiIndex).	`df.set_index(["Store", "Month"])`

Section 2 — Multiple-Choice Questions

Choose the correct option and circle the letter.

1. **What is the main data structure of NumPy?**
   - **A.** DataFrame **B.** Series
   - **C.** ndarray **D.** Dictionary
2. **Doing an operation on a whole array without a loop is called:**
   - **A.** Vectorization **B.** Broadcasting
   - **C.** Indexing **D.** Slicing
3. **Which Pandas structure is one-dimensional with labels?**
   - **A.** DataFrame **B.** Series
   - **C.** Panel **D.** ndarray

4. **Which one is a table with rows and columns?**
   A. Series **B.** ndarray
   C. Tuple **D.** DataFrame

5. **Which method gives summary statistics (mean, std, min, max)?**
   A. .dropna() **B.** .merge()
   C. .describe() **D.** .pivot()

6. **Which method removes rows that have missing values?**
   A. .dropna() **B.** .fillna()
   C. .isnull() **D.** .describe()

7. **Which method fills missing values with a chosen value?**
   A. .dropna() **B.** .fillna()
   C. .isnull() **D.** .sum()

8. **Which method checks for missing (NaN) values?**
   A. .sum() **B.** .mean()
   C. .isnull() **D.** .groupby()

9. **Which NumPy function reads array data from a text file?**
   A. pd.read_csv() **B.** np.loadtxt()
   C. np.plot() **D.** np.merge()

10. **What is a MultiIndex used for in Pandas?**
    A. Having more than one level of row labels **B.** Sorting one column
    C. Removing duplicate rows **D.** Drawing a chart

Section 3 — Fill in the Blanks

Word bank: ndarray vectorization DataFrame dropna() Hierarchical

1. NumPy's main data structure is called an _____.
2. Working on a full array at once, instead of using a loop, is called _____.
3. A Pandas _____ is a two-dimensional table with labels.
4. The _____ method removes rows that have missing values.
5. _____ indexing gives a DataFrame more than one level of row labels.

Section 4 — True or False

1. A Pandas Series is a two-dimensional data structure. **(True / False)**
2. All items in a NumPy array must have the same data type. **(True / False)**
3. The dropna() method fills missing values with a number. **(True / False)**
4. The describe() method gives mean, std, min and max. **(True / False)**
5. A DataFrame has rows and columns. **(True / False)**

Section 5 — Match the Following

Write the correct letter from Column B next to each item in Column A.

No.	Column A	Ans.	Column B
1	ndarray		a. Removes rows with missing values
2	Series		b. Table with rows and columns
3	DataFrame		c. Gives summary statistics
4	dropna()		d. NumPy's main array object
5	fillna()		e. One-dimensional labeled data
6	describe()		f. Fills missing values

Section 6 — Short-Answer Questions

Answer each question in 1–2 sentences.

1. **What is vectorization? Give one example.**

2. **Write one difference between a Series and a DataFrame.**

3. **Name two ways to handle missing data in Pandas.**

4. **What is hierarchical indexing (MultiIndex)? Give one example.**

5. **Give one example of an element-wise NumPy function.**

Section 7 — Practical Programs

Write and run each program in Python. Fill the blanks (___) using the hint.

Practical 1: Row and column averages (NumPy)

1. Create a 2-D array of exam scores with shape (5, 3).
2. Find the mean along axis=0 (columns) and axis=1 (rows).
3. Print both results.

Hint:

```
import numpy as np

scores = np.array([[70, 80, 90],
                   [60, 75, 85],
                   [88, 92, 79],
                   [55, 65, 70],
                   [95, 85, 90]])

print(scores.mean(axis=___))    # column averages
print(scores.mean(axis=___))    # row averages

# Tip: axis=0 goes down the rows, axis=1 goes across the columns
```

Expected output: *[73.6 79.4 82.8] and [80. 73.33 86.33 63.33 90.]*

Practical 2: Clean missing data (Pandas)

1. Create a DataFrame that has a few NaN values.
2. Count the missing values in each column using isnull().sum().
3. Fill the missing numbers with the column mean using fillna().

Hint:

```
import pandas as pd
import numpy as np

data = {"name":  ["Asha", "Ravi", "Meena", "Kiran"],
        "marks": [85, np.nan, 70, 90],
        "age":   [20, 21, np.nan, 22]}
df = pd.DataFrame(data)

print(df.isnull().___())                    # count missing values
df = df.fillna(df.___(numeric_only=True))   # fill with the column mean
print(df)
```

Expected output: *First: 1 missing in marks and 1 in age. Then a table with no NaN (Ravi's marks become 81.67).*

Practical 3: Summary statistics (Pandas)

1. Create a small DataFrame of sales figures.
2. Use describe() to see the summary statistics.
3. Print the mean and median of the "sales" column.

Hint:

```
import pandas as pd

df = pd.DataFrame({"store": ["A", "B", "C", "D", "E"],
                   "sales": [200, 350, 150, 400, 300]})

print(df.___())                  # summary table
print(df["sales"].mean())
print(df["sales"].___())      # middle value
```

Expected output: *A summary table (count, mean, std, min, 25%, 50%, 75%, max), then 280.0 and 300.0.*

Instructor Answer Key

Section 2 — MCQ

Q	Answer	Why
1	C	ndarray is NumPy's main array object.
2	A	Vectorization means working on the whole array at once.
3	B	A Series is 1-D with labels. A DataFrame is 2-D.
4	D	A DataFrame is a table with rows and columns.
5	C	.describe() gives count, mean, std, min, quartiles and max.
6	A	.dropna() removes rows that have NaN values.
7	B	.fillna() replaces NaN with a chosen value.
8	C	.isnull() marks which values are missing.
9	B	np.loadtxt() reads array data from a text file.
10	A	A MultiIndex gives more than one level of labels.

Section 3 — Fill in the Blanks

1. ndarray 2. vectorization 3. DataFrame 4. dropna() 5. Hierarchical

Section 4 — True or False

1. False 2. True 3. False 4. True 5. True

Section 5 — Match the Following

1 → d 2 → e 3 → b 4 → a 5 → f 6 → c

Section 6 — Short Answers (Key Points)

- Doing an operation on a whole array at once, with no loop. Example: arr * 2 doubles every item.
- A Series is one-dimensional (one column). A DataFrame is two-dimensional (a table).
- Remove them with dropna(), or replace them with fillna().
- Using more than one level of row labels. Example: Store and Month together.
- np.sqrt(arr), np.exp(arr) or np.abs(arr) (any one).

Section 7 — Practical Programs (Sample Solutions)

Practical 1

```
print(scores.mean(axis=0))    # [73.6 79.4 82.8]
print(scores.mean(axis=1))    # [80. 73.33 86.33 63.33 90.]
```

Practical 2

```
print(df.isnull().sum())
df = df.fillna(df.mean(numeric_only=True))
print(df)
```

Practical 3

```
print(df.describe())
print(df["sales"].mean())      # 280.0
print(df["sales"].median())    # 300.0
```